3F4D Water Monitor App: User Manual

Entry, automatic calculation, automatic writing to the Google Sheet and comparison with other sites

1. What the app does

3F4D Water Monitor is a smartphone app for recording the water level of the FP (Farmers' Practice) plot and the 3F4D+MD plot. When you save, the data are written automatically into the shared Google Sheet prepared by the administrator, through a script in the sheet (Google Apps Script). The dashboard shows the data kept in the sheet, not only the data on the phone. You can compare your site with the other sites in graphs.

Step	What happens
1. Enter	Students enter water level, irrigation, rain, fertilizer, pesticide and tiller counts in the app. Water level, irrigation and so on are calculated automatically.
2. Written automatically	When you save (for example "Save entry"), the data are written into the shared sheet a few seconds later. No pasting is needed.
3. Show and compare	Graphs and Compare show the data in the sheet. You can compare your site with the other sites in graphs.

Saving to the Google Sheet is required. "Save entry" cannot be pressed until the app is connected to the sheet.

2. Administrator setup (Excel template and sheet script)

The administrator of each site (for example a teacher, or a staff member of an NGO or SAAO) does this once at the start. Each site makes its own sheet.

1. **Make the sheet**: in the app, open More > For the administrator and download the template ATI_Monitor_Shared_TEMPLATE.xlsx with "Excel template". Upload it to Google Drive, open it and choose "File > Save as Google Sheets".
2. **Paste the script**: in the sheet open "Extensions > Apps Script". Delete the sample code that is already there, paste the script and press Save (the disk icon). You can copy the script with "Copy script" in the app (More > For the administrator); "Download script (.gs)" gives the file.
3. **Publish (deploy)**: press "Deploy > New deployment" at the top right. Use the gear icon to choose the type "Web app". Set "Execute as" to "Me" and "Who has access" to "Anyone", then press "Deploy".
4. The first time, an "Authorize access" screen appears. Choose your Google account. "Google hasn't verified this app" is shown, which is normal for a script you made yourself. Press "Advanced" > "Go to (project) (unsafe)" > "Allow".
5. **Web app URL**: when the deployment is done, a URL ending in "/exec" is shown. Copy it. (To see it later, use the sheet menu "3F4D Monitor > Show web app URL".)
6. **Connect the app**: in the app, open More > For the administrator and paste the URL into "Web app URL". Press "Test connection" and check that it says "Connected". Press "Create setup link" to make one link, and send that link to the users. When a user opens it, the URL is filled in automatically.

> **Note:** Only phones that opened the setup link (it contains the write key) can write to the sheet.

> Someone who knows only the URL can read but not write. Give the setup link to the users of your site only. If data are deleted by mistake, restore them with "File > Version history". If you change the script, update it with "Deploy > Manage deployments > Edit > New version". If the key is lost: sheet menu "3F4D Monitor > Reset write key", then make and send a new setup link.

What is in the sheet: On every save, the rows of that site are replaced (the sheet does not keep growing). The tab "Data" lists the daily results in a readable form, the tab "Sites" has one summary row per site, and "_Store" is a hidden tab used by the app. The script fills them in. Do not edit them by hand, and do not change the header rows or the tab names.

2.1 Comparing with other sites (compare links)

Each site uses its own Google Sheet (site A has one administrator and several users, site B another administrator and users, and so on). To compare sites, the administrators exchange "compare links". A compare link lets others read your data; they cannot change it.

1. **Make your link**: More > For the administrator > Compare with other sites > "Create compare link". Send the link to the administrators of other sites (WhatsApp, e-mail and so on).
2. **Add their links**: paste a compare link you received into "Add other sites" on the same screen and press "Add site" (or simply open the link). The added sites appear in Compare for every user of your site.
3. Press ✕ in the list to remove a site. Someone who is not a user of a site (for example a project member) can open a compare link to see Compare on that phone only.

3. How users use it

3.0 The five buttons at the bottom

Button	What it shows
Today	Whether today's entry is done, the latest water level, and "now" and "next" in the schedule. The first steps (0 to 3) also appear here.
Entry	Three parts: Water level (every day), Field records (fertilizer, pesticide, tillers) and TP / b (re-measurement).
Graphs	Graphs of water level, irrigation and saving of your site.
Compare	Comparison with the other sites added with compare links.
More	Setup, Schedule, Google Sheet & sharing, Help, For the administrator, demo.

3.1 The first time

- Open the "setup link" from the administrator in Chrome on the phone. This connects the app to the shared sheet. In Chrome, choose "Add to Home screen" to start the app from an icon.
- **More > Setup**: enter site name*, place*, sowing date*, seedling age at transplanting*, and TP and b* of FP and 3F4D+MD (* = required). While a required item is empty, or the app is not connected to the sheet, "Save entry" in the Entry tab cannot be pressed. The first screen, Today, shows whether steps 0 to 3 are done (0: the administrator prepares the sheet and the script, 1: connect to the sheet, 2: Setup, 3: first entry).
- **More > Schedule**: planned dates appear automatically. When you have done the work, open "Enter actual date / change plan" in that row and enter the actual date.

How to read m, TP and b: see the figure below. m = tape reading at the water surface, TP = tape reading at the top of the pipe, b = height of the pipe top above the ground (measured with another scale). Water level = m − TP + b. A negative value means the water is below the ground.

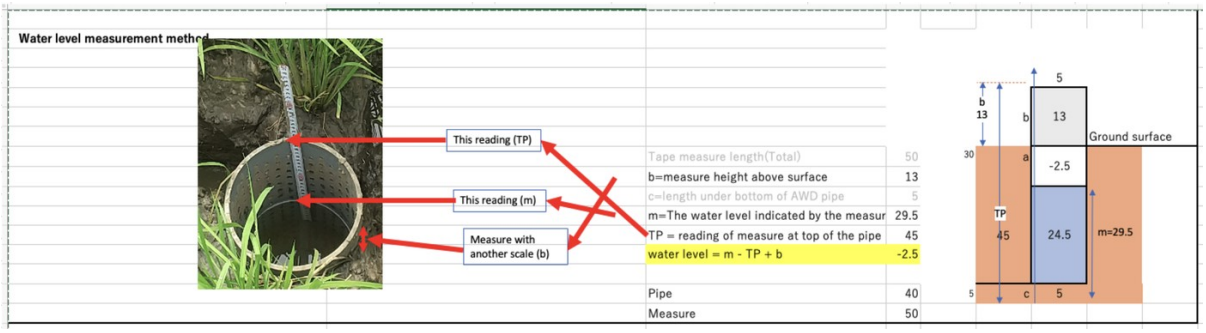


Figure: how to measure the water level (example: m = 29.5, TP = 45, b = 13 gives a water level of −2.5 cm)

3.2 Every day

1. **Today > Enter today's water level** (or Entry > Water level): enter the tape reading (m) of FP and 3F4D+MD. The water level (for example −2.5 cm, below the soil surface) appears at once under the box. The − and + buttons change m by 0.1 (starting from the last value when the box is empty). If you irrigated, choose "Yes" and enter the readings before and after irrigation; the irrigation (mm) also appears at once.
2. "Save entry" becomes active when all required items (*) are filled and the values are in range. Missing items are shown in red under the button.
3. After saving, the data are written into the sheet automatically a few seconds later. The line above the dashboard shows the state.

Message	Meaning and what to do
Not yet saved in the Google Sheet.	The sheet does not have it yet. It is sent automatically when the phone is online. You can also press "Send now".
Saved in Google Sheet	The sheet has it. This is the normal state.
No internet or no answer…	No internet, or the sheet does not answer. It is sent again when the connection is back. If this continues, ask the administrator to check the URL and the deployment.
The sheet script refused the data…	The sheet script did not accept the data. Tell the administrator.

You can also keep a photo of the pipe ("Add photo"). Photos stay on that phone only and are not sent to the sheet.

You can enter and save in the field without internet. The data stay on the phone for a while and are written into the sheet when the phone is online again.

Input	Required items
Setup	Site name, place, sowing date, seedling age, TP and b (both plots)
Entry (water level)	Date, m of FP, m of 3F4D+MD. If irrigation is "Yes": the readings before and after irrigation of that plot
Field records: fertilizer	Date, plot, at least one fertilizer amount (kg)
Field records: pesticide	Date, plot, type, chemical name
Field records: tillers	Date, plot, tiller counts of all 5 hills (whole numbers). The value used is the

Input	Required items
	mean of the 2nd, 3rd and 4th largest
Re-measure TP and b	Plot, date, TP, b

3.3 Correcting entries

- Water-level entry: press ✎ in the Entries list under Entry > Water level, change the values and press "Save entry" again (the same date is overwritten). ✕ deletes the day.
- Fertilizer, pesticide, tillers (Entry > Field records), TP and b (Entry > TP / b): press ✎ in the list, change the values and press "Update". Press "Cancel" to stop.
- After a correction the new version is also sent automatically. In the sheet, the newest version of the same site is used.

3.4 Continuing on a new phone

- Open the setup link. In More > Google Sheet & sharing > Restore from the sheet, type the site name (exactly as in the sheet) and press "Restore from sheet". The data in the sheet come back to that phone.

4. Comparing with other sites (Compare tab)

- The Compare tab reads your site and the other sites that your administrator added with compare links when the app opens (and again every 5 minutes). Press "Refresh from sheet" to read again. Tick the sites you want to see.
- Your own site is drawn with thick lines. The x axis is days after transplanting (DAT).
- Graphs: total irrigation and saving (FP vs 3F4D+MD), cumulative irrigation, water level, tiller number.
- "View" switches the display. "Overlay" draws all sites in the same graph. "Each site" shows a separate panel with water-level and cumulative-irrigation graphs for every site. It also works without data of your own (for example when an administrator looks at all sites).
- To see how Compare looks without connecting to the sheet, use More > Try with demo data (demo sites also appear in Compare). Nothing is sent in the demo.

5. Demo, backup and other points

- **Demo:** More > Try with demo data opens the app with example data. Nothing is saved or sent. Press "Exit demo" at the top to leave.
- **Backup:** The master copy is the sheet. The phone also keeps a copy until it has been sent. You may also use "Export file (JSON)" in More > Google Sheet & sharing from time to time.
- **View-only link:** "Create link" in More > Google Sheet & sharing makes a link that shows only the graphs.
- **Language:** The button at the top right switches English and Bengali. The Bengali is a draft and needs checking.

6. Notes and limits

- Each site name must be unique. The same name is treated as the same site.
- On every save, all data of that site are written into the sheet (not a difference). The script fills in the sheet; do not edit it by hand.

- Several people may enter data for one site on different phones. On every save the app merges with the sheet record by record (if two people change the same day, the later change is kept; a deleted record is deleted on the other phones too).
- Verified: the app's calculations equal the original Excel values; required-item checks, corrections, demo, writing to the script and restoring/loading from the sheet (with a mock in place of Google, including two simulated phones). The script itself (write, replace, read) was also tested on a mock sheet.
- Not verified: a real Google Sheet (the authorization screens, deployment, writing, reading), behaviour on real phones, correctness of the Bengali. Test with the first site before handing it out.

7. Sheet-only version (without the app)

For sites that prefer to enter data directly in a Google Sheet there is a sheet version (More > For the administrator > Sheet-only version). It does the same calculations as the app (water level, irrigation, cumulative irrigation, daily decrease, saving, tillers), the data checks, the schedule and the graphs, with formulas only. No script is needed. It has no comparison between sites.

1. Download "Sheet version (empty)", upload it to Google Drive and choose "Save as Google Sheets".
2. Share it as Editor with the users of the site. Each site uses its own copy ("File > Make a copy").
3. Same layout as the field irrigation-schedule workbook (dates across). Tabs: Today, Calendar (planned and actual dates of 3F4D+MD and FP), Data_Farmer (one column per day: water level, irrigation, work time, labour, condition, tillers and so on), Schedule_Farmer (planned and actual timeline), Setup, TP-b, Other_info (seedbed and main-field fertilizer and pesticide, N and P per 10a), Graphs, How_to_measure_water_level, pull-down list, Guide. Enter data in the yellow cells; do not change the grey (calculated) cells. A required cell that is empty turns red.

- "Sheet version (with demo data)" is the same sheet with example data. Its calculations were checked against the app (186 demo rows). How the graphs look after conversion to Google Sheets has not been checked yet.